Module - 2

Taylor's and maclaurin's for one variable.

Taylor's expansion

~~$y(x) = y(a) + (x-a)y_1(a)$~~.

$$y(x) = y + (x-a)y_1 + \frac{(x-a)^2}{2!} + \frac{(x-a)^3}{3!} + \cdots$$

maclaurian's expansion

$$y(x) = y(0) + xy_1(0) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(0) \cdots \frac{x^n}{n!}(0).$$

Exple for taylor's expansion

* Expand the term $\tan^{-1}(x)$ in powers of $(x-1)$ ~~is~~ given upto the term containing fourth degree.

→ Taylor's series expansion in powers of $(x-1)$ is given by

$$y(x) = y(1) + (x-1)y_1(1) + \frac{(x-1)^2}{2!}y_2(1) + \frac{(x-1)^3}{3!}y_3(1) + \frac{(x-1)^4}{4!}y_4(1) \longrightarrow ①$$

Using eq^n ① to findout

$y_1(1), y_2(1), y_3(1), y_4(1)$ only on diff w.r.t 'x'

Given $y = \tan^{-1}x \longrightarrow ②$

$y(x) = \tan^{-1}x$

$y(1) = \tan^{-1}x.$

$$\boxed{y = \pi/4}$$

$$y_1 = \frac{1}{1+x^2}$$

$$y_1(x) = \frac{1}{1+x^2} \longrightarrow ③$$

$$y_1(1) = \frac{1}{1+(1)^2}$$

$$y_1(1) = \frac{1}{2}$$

$(1+x^2)y_1 = 1$

Diff w r to 'x'

$(1+x^2)y_2 + 2xy_1 = 0 \longrightarrow (4)$

$(1+x^2)y_{2(x)} + 2xy_{1(x)} = 0$

$[1+(1)^2]y_{2(1)} + \cancel{2xy_{1(1)}}\ 2(1)\ \frac{1}{2} = 0$

$2y_{2(1)} = -1$

$\boxed{y_2 = -1/2} \longrightarrow$

Diff w r to x in (4)

$(1+x^2)y_3 + 4xy_{2(x)} + 2y_{1(x)} = 0 \longrightarrow (5)$

$(1+x^2)y_{3(x)} + 4xy_{2(x)} + 2y_{1(x)} = 0$

$[(1+(1)^2]y_3 + 4(1)(-1/2) + \cancel{2}\times\frac{1}{\cancel{2}} = 0$

$2y_3 + -2+1 = 0$

$2y_3 = +1$

$\boxed{y_3 = +1/2}$

Diff w r to x

$(1+x^2)y_4 + 2xy_3 + 4xy_3 + 4y_2 + 2y_2 = 0.$

$= (1+x^2)y_4 + 6xy_1 + 6y_2 = 0$

$(1+(1)^2)y_4 + 6(1)\frac{1}{2} + 6\times(-1/2) = 0$

$2y_4 + 3 + (-3) = 0$

$y_4 = 0$

Sub values in (1)

$y(x) = y(1) + (x-1)y_{1(1)} + \frac{(x-1)^2}{2!}y_2(1) + \frac{(x-1)^2}{3!}y_3(1)$

$+ \frac{(x-1)^2}{3!}y_4(1)$

$$y(x) = \tan^{-1}x = \pi/4 + (x-1)\tfrac{1}{2} + \frac{(x-1)^2}{2}\left(-\tfrac{1}{2}\right)$$
$$+ \frac{(x-1)^2}{3} \times \frac{1}{2} + \frac{(x-1)^4}{24}(0)$$

$$y(x) = \tan^{-1}x = \pi/4 + \frac{1}{2}\left[(x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{6}\right]$$

Exple for maclaurian's expansion

* Expand the term $\log(1+\sin x)$ in powers of 'x' in maclurian's theorem up to x^4.

Soln: Using maclorian's theorem upto the term containing x^4 is given by

$$y(x) = y(0) + x y_1(0) + \frac{x^2}{2!} y_2(0) + \frac{x^3}{3!} y_3(0) + \frac{x^4}{4!} y_4(0) \longrightarrow ①$$

And it is given that

$$y = \log(1+\sin x) \longrightarrow ②$$

$$\therefore\ y(x) = \log(1+\sin x)$$
$$y(0) = \log(1+\sin(0))$$
$$y(0) = \log(1)$$
$$\boxed{y(0) = 0}$$

Diff w r to x in ②

$$y_1 = \frac{1}{1+\sin x} \times \cos x$$

$$(1+\sin x) y_1 = \cos x \longrightarrow ③$$

$$(1+\sin(0)) y_1(x) = \cos x$$

$$(1+\sin(0)) y_1(0) = \cos 0$$

$$(1+0) y_1(0) = 1$$

$$\boxed{y_1(0) = 1}$$

Diff. w r to x in ③

$$(1+\sin x) y_2 + \cos x\, y_1 = -\sin x \longrightarrow ④$$

$$(1+\sin x) y_2(x) + \cos x\, y_1(x) = -\sin x$$

$$(1+\sin(0))\, y_2(0) + \cos(0)\, y_1(0) = -\sin(0)$$

$$(1+0)y_{2(0)} + (1)(1) = 0$$

$$\boxed{y_{2(0)} = 1}$$

Diff. w r to x in (4).

$$(1+\sin x)y_3 + y_2\cos x + y_2\cos x - \sin x\, y_1 = -\cos x$$

$$(1+\sin x)y_3 + 2y_2\cos x - \sin x\, y_1 = -\cos x \longrightarrow (5)$$

$$(1+\sin 0)y_{3(0)} + 2y_{2(0)}\cos(0) - \sin(0)y_{1(0)} = -\cos 0$$

$$(1+0)y_{3(0)} + 2y_{2(0)}(1) - 0 = -1$$

$$y_{3(0)} = -1+2(0)$$

$$\boxed{y_{3(0)} = +1}$$

Diff w r to x in (5)

$$(1+\sin x)y_4 + \cos x\, y_3 + 2[y_2(-\sin x) + y_3\cos x] + y_2\sin x$$
$$+ y_3 - [\cos x\, y_1 + y_2\cos x\sin x] = +\sin x$$

$$=(1+\sin x)y_{4(x)} + \cos x\, y_{3(x)} + 2[y_{2(x)}(-\sin x) + y_{3(x)}\cos x]$$
$$-\cos x\, y_{1(x)} - \sin x\, y_{2(x)} = \sin x$$

$$=(1+\sin 0)y_{4(0)} + \cos(0)y_{3(0)} + 2[y_{2(0)}(-\sin 0) + y_{3(0)}\cos 0]$$
$$-\cos(0)y_{1(0)} - \sin(0)y_{2(0)} = \sin(0)$$

$$=(1+0)y_{4(0)} + y_{3(0)}(1) + 2[y_{2(0)}(0) + y_{3(0)}(1)]$$
$$-(1)y_{1(0)} - 0.0)] = 0$$

$$= y_{4(0)} + 1 + 2[0+1] - (1) = 0$$

$$= \boxed{y_{4(0)} = -2}$$

Subtitute all the values in eqn (1)

$$y_{(x)} = y_{(0)} + xy_{1(0)} + \frac{x^2}{2!}y_{2(0)} + \frac{x^3}{3!}y_{3(0)} + \frac{x^4}{4!}y_{4(0)}$$

$$y(x) = 0 + x(1) + \frac{x^2}{2} \times (-1) + \frac{x^3}{3 \times 2 \times 1} \times 1 + \frac{x^4}{4 \times 3 \times 2 \times 1} (-2)$$

$$\boxed{y(x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12}}$$ hence the result.

Indeterminate forms

$\lim_{x \to 0} \frac{\sin x}{x} = 1$ $\lim_{x \to 0} \frac{x}{\sin x} = 1$ $\lim_{x \to 0} \frac{\tan x}{x} = 1$ $\lim_{x \to 0} \frac{x}{\tan x} = 1$

Exple:

* $\lim_{x \to 0} \frac{\tan x - x}{x^2 \tan x}$

Soln: Let $K = \lim_{x \to 0} \frac{\tan x - x}{x^2 \tan x}$ ---- $\left(\frac{0}{0}\right)$

i.e., $= \lim_{x \to 0} \frac{\tan x - x}{\frac{x^2 \tan x}{x} \cdot x}$

$= \lim_{x \to 0} \frac{\tan x - x}{x^3} \times \frac{x}{\tan x}$ $\left[\because \frac{x}{\tan x} = 1\right]$

$= \lim_{x \to 0} \frac{\tan x - x}{x^3} \times 1$ --- $\left(\frac{0}{0}\right)$

Applying L' Hospital's rule,

$$K = \lim_{x \to 0} \frac{\sec^2 x - 1}{3x^2}$$

$$K = \lim_{x \to 0} \frac{\tan^2 x}{3x^2}$$

$$K = \frac{1}{3} \lim_{x \to 0} \left(\frac{\tan x}{x}\right)^2$$

$$K = \frac{1}{3} \cdot 1$$

$$\boxed{K = \frac{1}{3}}$$

Partial differentiation

$u = f(x, y)$

$u_x = \frac{\partial u}{\partial x} \qquad u_y = \frac{\partial u}{\partial y}$

$u_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial x^2}$

$u_{yy} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial y^2}$

$u_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial y \partial x}$

$u_{xy} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial^2 u}{\partial x \partial y}$

$\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$

$u_{yx} = u_{xy}$ //

Exple

* If $u = \log\sqrt{x^2+y^2+z^2}$, show that $(x^2+y^2+z^2)\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = 1$

Sol^n using partial differentiation

Given $u = \log\sqrt{x^2+y^2+z^2}$

$u = \frac{1}{2}\log(x^2+y^2+z^2)$

$\frac{\partial u}{\partial x} = \frac{1}{\cancel{2}} \cdot \frac{1}{x^2+y^2+z^2} \times \cancel{2}x = \frac{x}{x^2+y^2+z^2}$

$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2+z^2}\right)$

$= \frac{(x^2+y^2+z^2)1 - x \cdot 2x}{(x^2+y^2+z^2)^2}$

$\frac{\partial^2 u}{\partial x^2} = \frac{y^2+z^2-x^2}{(x^2+y^2+z^2)^2} \longrightarrow (1)$

Illy
$$\frac{\partial u}{\partial y} = \frac{1}{\cancel{2}} \frac{1}{x^2+y^2+z^2} \times \cancel{2}y = \frac{y}{x^2+y^2+z^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{(x^2+y^2+z^2)1 - y \cdot 2y}{(x^2+y^2+z^2)^2}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{x^2+z^2-y^2}{(x^2+y^2+z^2)^2} \longrightarrow (2)$$

$$\frac{\partial u}{\partial z} = \frac{1}{\cancel{2}} \frac{1}{x^2+y^2+z^2} \cdot \cancel{2}z = \frac{z}{x^2+y^2+z^2}$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{(x^2+y^2+z^2) \cdot 1 - z \cdot 2z}{(x^2+y^2+z^2)^2}$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{x^2+y^2-z^2}{(x^2+y^2+z^2)^2} \longrightarrow (3)$$

Adding (1), (2) & (3) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{y^2+z^2-x^2}{(x^2+y^2+z^2)^2} + \frac{x^2+z^2-y^2}{(x^2+y^2+z^2)^2} + \frac{x^2+y^2-z^2}{(x^2+y^2+z^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{y^2+z^2-\cancel{x^2}+\cancel{x^2}+\cancel{z^2}-\cancel{y^2}+x^2+\cancel{y^2}-\cancel{z^2}}{(x^2+y^2+z^2)^2}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{\cancel{y^2+z^2+x^2}}{(x^2+y^2+z^2)^{\cancel{2}}}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{(x^2+y^2+z^2)}$$

by cross multiplication

$$(x^2+y^2+z^2)\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = 1$$

Total derivative - differentiation of Composite fux^ns

If $u = f(x,y)$ then the total differential of u is defined as

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \longrightarrow ①$$

1. Total derivative Rule.

If $u = f(x,y)$ where $x = x(t)$ & $y = y(t)$ then u is a Composite function of the single Variable t.

Then eq^n ① becomes

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} \longrightarrow ②$$

2. Chain rule

If $u = f(x,y)$ where $x = x(r,s)$ & $y = y(r,s)$ then u is a Composite function of two independent Variables (r,s).

$\therefore$ eq^n ① becomes

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} + \frac{\partial y}{\partial r}$$

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial s}$$

Exple :

* find the total derivative $z = xy^2 + x^2y$ where $x = at$, $y = 2at$

Sol^n given $z = xy^2 + x^2y$; $x = at$, $y = 2at$

$\{z \longrightarrow (x,y) \longrightarrow t\} \Rightarrow z \longrightarrow t$ & $\frac{dz}{dt}$ is the total derivative.

$$\therefore \frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial t}$$

$x = at$

$y = 2at$

$$= (y^2 + 2xy)a + (2xy + x^2)2a$$

$$= [(2at)^2 + 2(at)(2at)]a + [2(at)(2at) + (at)^2]2a$$

$$= (4a^2t^2 + 4a^2t^2)a + (4a^2t^2 + a^2t^2)2a$$

$$= (8a^2t^2)a + (5a^2t^2)2a$$

$$= 8a^3t^2 + 10a^3t^2$$

$$= 18a^3t^2$$

$$\therefore \frac{dz}{dt} = 18a^3t^2 \longrightarrow \text{Hence the result.}$$

Jacobians

* Let u and v be functions of two independent variables x and y. The jacobian (J) of u & v with respect to x & y is represented as.

$$J\left(\frac{u,v}{x,y}\right) \text{ or } \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

Hly if u, v, w are functions of three independent variables x, y, z then

$$J\left(\frac{u,v,w}{x,y,z}\right) \text{ or } \frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

Exple

* find the jacobian of u, v, w w.r.t x, y, z given that $u = x + y + z$, $v = y + z$; $w = z$

Soln We have to find $J = \dfrac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial z}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$

$u = x+y+z \,,\ v = y+z \,,\ w = z$

$$\therefore \quad J = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

on expanding by the last row we get

$J = 1\,(1-0) = 1$

$\therefore \ \boxed{J = 1}$

Maxima and minima for a function of two Variables.

Definition: A function $f(x,y)$ is said to have a maximum value at a Point (a,b) if there exists a neighbourhood of the point (a,b) say $(a+h,\ b+k)$, h and k are small such that $f(a,b) > f(a+h,\ b+k)$

Ily if $f(a,b) < f(a+h, b+k)$ then $f(x,y)$ is said to have a minimum value at (a,b). Also $f(a,b)$ is said to be an extreme value of $f(x,y)$ if it is a maximum value or a minimum value.

Working procedure

* We have to first find the stationary points (x,y) such that $f_x = 0$ and $f_y = 0$
* We then find the second order partial derivatives $A = f_{xx}$, $B = f_{xy}$, $C = f_{yy}$.

we evaluate these at all the stitionary points and also compute the corresponding value of $AC - B^2$.

* (a) A stationary point (x_0, y_0) is a maximum value. $AC - B^2 > 0$ & $A < 0$. $f(x_0, y_0)$ is a maximum value

(b) A stationary point (x_1, y_1) is a minimum value $AC - B^2 > 0$ & $A > 0$. $f(x_1, y_1)$ is a minimum value.

Exple

* Find the extreme values of the function $f(x,y) = x^3 + y^3 - 3x - 12y + 20$.

Sol^n $f(x,y) = x^3 + y^3 - 3x - 12y + 20$.

$f_x = 3x^2 - 3$ $\qquad f_y = 3y^2 - 12$.

We shall find points (x,y) such that $f_x = 0$ & $f_y = 0$

i.e $3x^2 - 3 = 0$ $\qquad 3y^2 - 12 = 0$

$x^2 - 1 = 0$ $\qquad y^2 - 4 = 0$

$x^2 = 1$ $\qquad y^2 = 4$

$x = \pm 1$ $\qquad y = \pm 2$.

$\therefore$ $(1,2), (1,-2), (-1,2), (-1,-2)$ are the stionary points

Let $A = f_{xx} = 6x$ $\qquad \boxed{\begin{array}{l} f_x = 3x^2 - 3 \\ f_y = 3y^2 - 12 \end{array}}$

$B = f_{xy} = 0$

$C = f_{yy} = 6y$

	$(1,2)$	$(1,-2)$	$(-1,2)$	$(-1,-2)$
$A = 6x$	$6 > 0$	6	-6	$-6 < 0$
$B = 0$	0	0	0	0
$C = 6y$	12	-12	12	-12
$AC - B^2$	$72 > 0$	$-72 < 0$	$-72 > 0$	$72 < 0$
Conclusion	minimum point	Saddle point	Saddle point	maximum point.

Maximum value of $f(x,y)$ is $\quad f(x,y) = x^3+y^3-3x-12y+2$

$$f(-1,-2) = (-1)^2 + (-2)^3 - 3(-1) - 12(-2) + 20$$

$$= -1 + 8 + 3 + 24 + 20$$

$$= 38$$

Minimum value of $f(x,y)$ is.

$$f(1,2) = (1)^3 + (2)^3 - 3(1) - 12(2) + 20$$

$$= 1 + 8 - 3 - 24 + 20$$

$$= 2 \,/\!/$$